Comprehensive Research Dossier: Casimir Funk and the Biochemical Foundations of Vitaminology

A) ANCHOR SCENES

SCENE #1: The Linguistic Resistance at the Lister Institute Summary: Casimir Funk, a twenty-seven-year-old Polish biochemist, finds his revolutionary conceptual framework for "vitamines" suppressed by the senior editorial authority of the Lister Institute, forcing a clandestine publication route to introduce the term to the world. Date/Place: Late 1911 to June 1912; The Lister Institute of Preventive Medicine, London, United Kingdom. People: Casimir Funk (Researcher); Dr. Charles James Martin (Director of the Lister Institute); Ludwik Rajchman (Publisher of the Journal of State Medicine). Concrete details: The laboratory at the Lister Institute was characterized as "rather primitive," where Funk conducted his fractionation experiments often with only the assistance of a non-scientific "servant".[1] Having isolated a substance from rice polishings that appeared to cure polyneuritis in pigeons, Funk prepared a manuscript for *The Lancet* titled "Experiments on the causation of Beri-Beri".[2] He had coined the term "vitamine"—a portmanteau of *vita* (life) and *amine* (nitrogenous compound)—to describe these essential trace elements.[3] However, the institutional leadership, led by Charles Martin, viewed the term as scientifically "speculative" because the chemical nature of the substances had not been definitively proven as amines.[2] Consequently, the word was purged from his 1911 *Lancet* paper.[2] Frustrated, Funk sought an alternative venue. He leveraged his friendship with Ludwik Rajchman, who oversaw the *Journal of State Medicine*, to publish "The Etiology of Deficiency Diseases" in 1912.[5] In this paper, the word "vitamine" appeared in print for the first time, marking a defiance of institutional gatekeeping.[2] Why it matters: This scene illustrates the friction between radical theoretical innovation and the conservative demands of empirical proof. It established the nomenclature that would define 20th-century nutrition and demonstrated Funk's commitment to his unifying theory over institutional approval. Evidence quality: High. Multiple secondary sources and institutional histories document the specific rejection by the Lister Institute and the subsequent publication in the *Journal of State Medicine*.[2] Myth risk: Medium. While the institutional rejection is factually documented, the specific emotional tension between Funk and Martin is often dramatized in later biographies like Harrow's. Sources: [1]

SCENE #2: The Great Fractionation and the Niacin Mirage Summary: Through a massive effort of chemical separation, Funk isolates what he believes to be the pure anti-beriberi factor, unaware that he has actually discovered a different vitamin (Niacin) while his target (Thiamine) remains an elusive contaminant. Date/Place: 1911–1912; Lister Institute, London. People: Casimir Funk; Experimental Pigeons. Concrete details: Funk began with massive

quantities of rice polishings, the cortical layers removed during the milling of white rice.[2] Using a method of "fracturing," he divided the polishings into Fraction A and Fraction B.[2] The workspace was filled with the smell of yeast and the residue of thousands of rice husks.[7] He observed that pigeons suffering from polyneuritis—the avian equivalent of human beriberi—died when fed Fraction A but recovered miraculously when fed minute quantities of Fraction B.[2] Within Fraction B, he eventually isolated tiny, needle-like crystals.[9] He believed these were the pure "vitamine" of beriberi.[9] However, modern chemical retrospection reveals these crystals were actually nicotinic acid (niacin), which was contaminated with just enough thiamine to cure the birds.[9] When the pure nicotinic acid failed to consistently cure beriberi in subsequent trials, Funk was forced to set it aside, essentially "missing" the discovery of the cure for pellagra while chasing thiamine.[9] Why it matters: This scene encapsulates the "near-miss" nature of early biochemistry. Funk's failure to isolate pure thiamine led him to the correct theoretical conclusion that *multiple* vitamins must exist, even if his chemical isolation was imperfect. Evidence quality: High. The chemical identification of his 1912 isolate as nicotinic acid is a verified milestone in nutritional history.[9] Myth risk: Low. The chemical discrepancy is a well-analyzed historical fact. Sources: [2]

SCENE #3: The "White Eagle" Return: Establishing Polish Biochemistry Summary: Following the restoration of Polish independence, Funk returns to Warsaw as a national scientific hero, utilizing his international prestige and his own personal wealth to establish the country's first modern biochemical infrastructure. Date/Place: 1923–1927; State Institute of Hygiene (PZH), Warsaw, Poland. People: Casimir Funk; Ludwik Rajchman (Director of PZH); Polish medical students. Concrete details: Funk returned to a Poland newly emerged from the partitions, sponsored by the Rockefeller Foundation.[12] He was appointed Chief of the Department of Biochemistry at the State Institute of Hygiene.[12] The laboratory conditions were initially subpar, requiring significant modernization.[12] Funk did not merely direct research; he personally "invested his own funds" to ensure the facility could function during the economic volatility of the early Polish Republic.[14] One of his most significant operational achievements was the initiation of domestic insulin production in 1924, making Poland the third country in Europe to manufacture the hormone.[14] He worked against a backdrop of "political unrest," yet he maintained a rigorous schedule of scientific diplomacy, traveling to conferences in Copenhagen and Bucharest to represent Polish science.[5] Why it matters: This transition from theoretical research in London to institutional building in Warsaw showcases Funk’s sense of national identity and his belief that biochemistry was a tool for public health and state sovereignty. Evidence quality: Strong. Institutional records from the PZH and the Rockefeller Foundation confirm his tenure and the insulin production milestones.[12] Myth risk: Low. The dates and specific achievements in Warsaw are well-recorded in Polish scientific annals. Sources: [5]

SCENE #4: The Founding of "Casa Biochemica" in Paris Summary: Driven out of Poland by political instability and funding cuts, Funk establishes a private, commercially viable research

institute in France, attempting to prove that high-level research can be self-sustaining outside of academia. Date/Place: 1927–1939; Rueil-Malmaison, Paris, France. People: Casimir Funk; Industrial partners from the Roussel Company. Concrete details: In 1928, citing the increasing "political unrest" and the difficulty of maintaining state funding in Poland, Funk relocated to Paris.[12] He founded a private laboratory named "Casa Biochemica".[12] This was not a purely academic endeavor; the lab produced biochemical products to fund its research mission.[12] Funk worked as a consultant for the Roussel Company during this period (1927–1936), focusing heavily on the biochemistry of sex hormones.[12] It was here that he performed pioneering work on the synthesis of androsterone and continued his research into the relationship between nutrition and cancer.[5] The laboratory became a hub for "travelling scientists," reflecting Funk's own polyglot and international career.[5] Why it matters: This scene highlights the entrepreneurial necessity of early 20th-century scientists. Funk's decision to go private was a precursor to modern biotechnology firms, where commercial success fueled basic research into hormones and vitamins. Evidence quality: Medium. While the existence and purpose of Casa Biochemica are cited in biographies and finding aids, specific lab notebooks from this era are less publicly accessible than his Lister Institute work.[5] Myth risk: Medium. The specific "political unrest" that caused his departure from Poland is often cited vaguely without detailing specific conflicts. Sources: [5]

SCENE #5: The Nobel Snub of 1929 Summary: The scientific community gathers to celebrate the discovery of vitamins by awarding the Nobel Prize to Christiaan Eijkman and Frederick Gowland Hopkins, notably omitting Funk, whose unifying theory and naming provided the field's essential framework. Date/Place: October 1929; Stockholm, Sweden / New York City (where Funk was living). People: Christiaan Eijkman (Nobel Laureate); Frederick Gowland Hopkins (Nobel Laureate); Casimir Funk (Nominee). Concrete details: The Nobel Committee awarded the Prize in Physiology or Medicine to Eijkman "for his discovery of the antineuritic vitamin" and to Hopkins "for his discovery of the growth-stimulating vitamins".[10] Funk had been nominated for the prize four times (1914, 1925, 1926, and 1946).[13] Critics and supporters alike pointed out that while Eijkman observed the phenomenon and Hopkins provided the growth data, it was Funk who synthesized these disparate observations into the "vitamine hypothesis".[15] Funk's reaction was reportedly one of resigned humility; he later noted that "considering the circumstances... there is no single person who deserves this distinction".[15] This omission remains one of the most significant priority disputes in the history of biochemistry.[15] Why it matters: This scene defines the historical memory of Casimir Funk as a "scientist's scientist"—someone whose conceptual contributions were so pervasive they became taken for granted, even as the formal accolades were bestowed upon those who provided the initial data points. Evidence quality: High. Nobel nomination records are public after 50 years and confirm Funk's multiple nominations and the eventual winners.[13] Myth risk: Low. The details of the 1929 award and the surrounding controversy are well-documented in history-of-science literature. Sources: [10]

B) DIRECT QUOTES FROM THE SCIENTIST

1. "The deficient substances, which are of the nature of organic bases we will call 'vitamines'; and we will speak of a beri-beri or scurvy vitamine, which means a substance preventing the special disease." — Published in "The Etiology of Deficiency Diseases," *Journal of State Medicine*, 1912.[1]
2. "Lack of a particular vitamin leads eventually to a particular nutritional disease." — Written in *Vitamin and Mineral Therapy*, 1936.[12]
3. "Considering the circumstances of the discovery of vitamins, there is no single person who deserves this distinction [the Nobel Prize]." — Retrospective comment on the 1929 Nobel Prize controversy.[15]
4. "The science [of nutrition] was 'in its infancy,' it has not been so thoroughly investigated as to justify anyone laying down [rigid rules]." — Warning against early 20th-century nutritional dogmatism, c. 1915–1920.[20]
5. "A monotonous diet ought to be avoided, because in this case a deficient food is made use of for long periods and prepares the ground for the outbreak of the deficiency diseases." — *The Etiology of the Deficiency Diseases*, p. 366, 1912.[7]
6. "It is because on better acquaintance they [vitamines] have compelled us to reform our ideas on nutrition of both adults and babies and pick out our foods from a new angle, that we accord them the attention they demand and deserve." — *The Vitamines*, 1922 English Edition Preface.[21]

C) HUMAN REALITY DETAILS

- **Early Orthopedic Struggles:** Born with hip dysplasia in Warsaw, Funk was sent to a clinic in Germany as a young child for long-term treatment. This early isolation and convalescence are credited with fostering his intense focus and fluency in German.[6]
- **The Multilingual Scholar:** Funk was a true internationalist, fluent in Polish, Russian, German, French, and English, which allowed him to publish and collaborate seamlessly across five different national scientific traditions.[5]
- **Academic Prodigy:** He completed his entire doctoral program and earned a PhD in chemistry from the University of Bern by the age of 20 in 1904.[6]
- **Parental Lab Assistants:** During his Beit Fellowship in London (1913), Funk's parents, Jacques and Gustawa, joined him. His father, a noted dermatologist, personally assisted in data collection and literature abstraction to support his son's research.[1]
- **Financial Risk-Taker:** During his tenure in Warsaw (1923–1927), Funk utilized his own personal savings to fund the initial production runs of insulin when state funding was delayed by the post-war economic crisis.[14]
- **Minimalist Lab Culture:** At the Lister Institute, Funk was known for working under "rather primitive" conditions, frequently accompanied only by a non-scientific "servant" who assisted with the manual labor of processing hundreds of pounds of rice hulls.[1]
- **Professional Mentorship:** Funk mentored Jack Drummond at the London Cancer

Hospital; Drummond would later become the chief nutrition advisor to the British government during the critical rationing years of World War II.[1]

- **A Personal Library of Evolution:** Despite his frequent migrations, Funk maintained a prized personal collection that included two rare works of Charles Darwin translated into Polish, reflecting his deep intellectual roots in evolutionary biology.[24]
- **The Transatlantic Pivot:** He became a U.S. citizen in 1920 but spent much of the interwar period in Europe, only returning permanently to New York in 1939 as the Nazi threat made European Jewish life untenable.[12]
- **Quiet Finality in Albany:** After a career in the world's most bustling scientific capitals, Funk spent his final years in the quieter environment of Albany, New York, where he continued to correspond with the U.S. Vitamin Corporation until his death at age 83.[3]

D) SCIENCE CONNECTION (Intro Biochem Bridge)

Best-fit Topic: Metabolic Coenzymes and the Pyruvate Dehydrogenase Complex.

Precise Contribution: Casimir Funk proposed the "vitamine hypothesis," theorizing that specific organic nitrogenous compounds (amines) are essential dietary micronutrients whose absence causes metabolic "biochemical lesions" resulting in deficiency diseases.[4]

Bridge Explanation: In introductory biochemistry, students study how enzymes require non-protein "helpers" called cofactors to function. Thiamine (Vitamin B1), the substance Funk spent decades trying to isolate, is the biological precursor for Thiamine Pyrophosphate (TPP).[25] TPP is an essential coenzyme for the Pyruvate Dehydrogenase Complex, which facilitates the oxidative decarboxylation of pyruvate into Acetyl-CoA, the entry point for the Citric Acid Cycle.[25] Without B1, the body cannot process carbohydrates into ATP, leading to the cellular energy crisis observed in beriberi.[8]

Chemical Reaction Context:

$$Pyruvate + CoA + NAD^+ \xrightarrow{TPP, Lip, FAD} Acetyl - CoA + CO_2 + NADH + H^+$$

Funk's intuition that these substances might be "transformed into ferments [enzymes]" was a prescient precursor to our modern understanding of TPP as a prosthetic group.[1]

E) COMPLICATIONS, DISPUTES, OR CONTROVERSIES

The "Amine" Misnomer and the Final "e"

- **What's agreed:** Funk named the substances "vitamines" based on his belief that they were all nitrogen-containing organic bases (amines).[3]
- **What's disputed:** In 1920, Cecil Drummond pointed out that many "vitamines" (such as

Vitamin A) did not contain nitrogen or amine groups.[10] This led to the scientific community dropping the final "e" to create the neutral term "vitamin," a move Funk initially resisted but eventually accepted as the chemical diversity of vitamins became undeniable.[15]

The Priority Dispute: Eijkman vs. Hopkins vs. Funk

- **What's agreed:** Eijkman observed the effect in chickens; Hopkins proved "accessory factors" were needed for growth; Funk provided the name and the unifying theory.[9]
- **What's disputed:** Who is the true "discoverer"? Eijkman initially believed beriberi was caused by a toxin in the rice endosperm that the bran neutralized (the "antidote theory"), which was fundamentally incorrect.[17] Hopkins provided data but no overarching disease-prevention theory. Funk provided the theory but failed to isolate a *pure* sample of B1 in 1912.[9] The Nobel Committee chose data (Eijkman/Hopkins) over conceptual synthesis (Funk).

The Niacin/Thiamine Confusion of 1912

- **What's agreed:** The crystalline needles Funk isolated in 1912 were nicotinic acid (Niacin/B3).[9]
- **What's disputed:** Many contemporaries viewed Funk's isolation as a "failure" because it did not cure beriberi reliably. However, modern historians argue it was an accidental second discovery—Funk had isolated the cure for pellagra years before it was recognized, but because he was looking for the cure for beriberi, he discarded the most significant chemical result of his early career.[11]

F) "NOT ON WIKIPEDIA" GOLD

1. **DOPA Synthesis:** In 1911, while a scholar at the Lister Institute, Funk was the first to synthesize 3,4-dihydroxyphenylalanine (DOPA). This achievement was a major milestone in organic chemistry, decades before DOPA's role in treating Parkinson's disease was understood.[1]
2. **Early Nutritional Oncology:** As early as 1913, while working at the London Cancer Hospital Research Institute, Funk conducted experiments suggesting that specific vitamins could influence the speed of tumor growth, marking the birth of the field of nutritional oncology.[12]
3. **The "Yeast Beast" and WWI:** During World War I, Funk collaborated with H.A. Metz & Co. to produce "Oscodal," a concentrated vitamin supplement. This work led to the deployment of brewer's yeast (Marmite) as a prophylactic against beriberi for British troops in Mesopotamia in 1916.[14]
4. **Estrogen Synthesis at 20:** Funk's doctoral dissertation at the University of Bern (1904) involved the successful synthesis of estrogen, establishing his expertise in hormone biochemistry long before he pivot to nutritional deficiency studies.[6]

5. **The "Servant" Assistant:** Archival records from the Lister Institute reveal that Funk's breakthrough research was conducted without a single trained research assistant. His only help was a laboratory servant who assisted with the arduous physical task of boiling and filtering hundreds of pounds of rice polishings.[1]

G) SOURCES CONSULTED

- **Primary Source:** Funk, Casimir. *The Vitamines*. Baltimore: Williams and Wilkins, 1922. (English translation of the original 1912/1913 German text). [5]
- **Biographical Memoir:** Harrow, Benjamin. *Casimir Funk: Pioneer in Vitamins and Hormones*. New York: Dodd, Mead & Company, 1955. (Includes personal correspondence and lab recollections). [1]
- **Scholarly Journal:** Piro, A., et al. "Casimir Funk: His Discovery of the Vitamins and Their Deficiency Disorders." *Annals of Nutrition and Metabolism*, 57(2), 2010. [4]
- **Institutional Archive:** "Casimir Funk Collection Finding Aid," Huntington Library, San Marino, CA. (Contains drafts of his unpublished autobiography). [16]
- **Historical Monograph:** Carpenter, Kenneth J. *Beriberi, White Rice, and Vitamin B1: A Disease, a Cause, and a Cure*. University of California Press, 2000. [8]
- **Commemorative Article:** Berger, S. "Kazimierz (Casimir) Funk – Pioneer in Vitaminology." *Polish Journal of Food and Nutrition Sciences*, 63(4), 2013. [5]
- **Priority Review:** Semba, R.D. "The Discovery of the Vitamins." *International Journal for Vitamin and Nutrition Research*, 82, 2012. [10]

H) FLAGGED "MYTH-RISK" DETAILS

- **Claim:** Funk discovered Vitamin B1 in a sudden "eureka" moment.
  - **Reality:** The process was a grinding, multi-year fractionation effort that actually resulted in an impure sample of Niacin (B3) contaminated with B1. There was no single moment of clear, pure isolation in 1911.[8]
- **Claim:** Funk was the first to propose that food contains essential trace factors.
  - **Reality:** Scientists like Lunin (1881) and Pekelharing (1905) had observed that "synthetic" diets failed where natural milk succeeded. Funk's unique contribution was not the observation, but the **unifying theory** that linked these factors to specific diseases like scurvy and pellagra.[15]
- **Claim:** The name "vitamine" was immediately and universally celebrated.
  - **Reality:** It was initially suppressed by the Lister Institute and *The Lancet* for being too speculative, and it was only through a secondary, smaller journal that Funk managed to publish it.[2]

LENS RECOMMENDATION

Lens	Score (0-3)	Justification
Eureka	1	No singular "pure" discovery moment; his 1912 isolate was actually the "wrong" vitamin (Niacin).
Failure	2	The 1929 Nobel snub is significant, but Funk's career remained highly productive and prestigious.
Mentor	1	He worked with giants (Fischer, Bertrand), but the evidence of a deep, life-changing mentorship is thin.
Controversy	3	**Primary Lens.** The nomenclature war with the Lister Institute and the lifelong priority dispute are the most vivid, well-documented narrative drivers.
Human Detail	3	**Backup Lens.** The hip dysplasia, the parents in the lab, and the multilingual migrations provide a rich, sympathetic "life story."
Last Experiment	1	Retirement in Albany is well-documented but lacks a dramatic "final scene" quality.

Primary Recommendation: The Controversy Lens

- **Why it fits:** The most compelling scenes (Scene #1 and Scene #5) revolve around Funk

fighting for recognition and the right to name his own discovery. The conflict between the "Speculative Pole" and the "Conservative British Establishment" provides a natural narrative tension.

- **Best Opening Scene:** Scene #1 (The Linguistic Resistance). Starting with the censorship of the word "vitamine" immediately hooks the reader into a story of intellectual bravery.
- **Best Quote:** Quote #1 (The 1912 definition). It marks the exact moment the field was born, despite institutional pushback.
- **Caution:** Avoid portraying Funk as a victim; emphasize that his "controversial" nomenclature actually won the global war for the public's imagination, even if it lost the local war with the Nobel Committee.

Backup Recommendation: The Human Detail Lens

- **Why it fits:** Funk's life is a "scientific odyssey." From his childhood orthopedic struggles in Poland to his rapid-fire PhD in Switzerland and his global migrations, the focus on his personal persistence makes the science more accessible to a general student audience. Use Scene #3 (The Return to Warsaw) as the emotional core.

Works cited

1. CASIMIR FUNK, accessed January 26, 2026, https://nut.sourceforge.net/funk.pdf
2. Casimir Funk and a Century of Vitamins | PaulingBlog, accessed January 26, 2026, https://paulingblog.wordpress.com/2011/06/15/casimir-funk-and-a-century-of-vitamins/
3. Funk, Casimir (Polish Biochemist) – Study Guide | StudyGuides.com, accessed January 26, 2026, https://studyguides.com/topics/cmk19s4stague01d5knsh9tcj
4. Casimir Funk: his discovery of the vitamins and their deficiency disorders - PubMed, accessed January 26, 2026, https://pubmed.ncbi.nlm.nih.gov/20805686/
5. Kazimierz (Casimir) Funk – Pioneer in Vitaminology – 101 ..., accessed January 26, 2026, https://journal.pan.olsztyn.pl/pdf-98368-31138?filename=31138.pdf
6. Kazimierz Funk – The Elixir of Life | Article - Culture.pl, accessed January 26, 2026, https://culture.pl/en/article/kazimierz-funk-the-elixir-of-life
7. Political ecology and the industrial food system - Squarespace, accessed January 26, 2026, https://static1.squarespace.com/static/57195cec8259b5e997974243/t/612a9c5a9e9d2a7651dfbadf/1630182491931/Political+ecology+and+industrial+food+PBH.pdf
8. Beriberi, White Rice, and Vitamin B : A Disease, a Cause, and a Cure, accessed January 26, 2026, http://111.68.96.114:8088/get/pdf/Beriberi%2C%20White%20Rice%2C%20and%20Vitamin%20B_%20A%20Disease%2C%20a%20Cause%2C%20and%20a%20Cure%20-%20Kenneth%20J.%20Carpenter_6077.pdf
9. Hopkins Postulates the Presence of Vitamins | Research Starters - EBSCO, accessed January 26, 2026, https://www.ebsco.com/research-starters/history/hopkins-postulates-presence-vitamins

10. What Is a Vitamin? Towards a Contemporary Definition - PMC - PubMed Central, accessed January 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12735764/
11. Niacin History - News-Medical.Net, accessed January 26, 2026, https://www.news-medical.net/health/Niacin-History.aspx
12. Casimir Funk | Encyclopedia.com, accessed January 26, 2026, https://www.encyclopedia.com/people/medicine/medicine-biographies/casimir-funk
13. (PDF) Kazimierz Funk: Polish vitamins' discoverer - ResearchGate, accessed January 26, 2026, https://www.researchgate.net/publication/381010596_KAZIMIERZ_FUNK_POLISH_VITAMINS'_DISCOVERER
14. Funk Kazimierz | Virtual Shtetl, accessed January 26, 2026, https://sztetl.org.pl/en/biographies/196575-funk-kazimierz
15. Casimer Funk, Nonconformist Nomenclature, and Networks Surrounding the Discovery of Vitamins - ResearchGate, accessed January 26, 2026, https://www.researchgate.net/publication/236967454_Casimer_Funk_Nonconformist_Nomenclature_and_Networks_Surrounding_the_Discovery_of_Vitamins
16. Casimir Funk Collection., 1896-1984, bulk 1939-1966 - OAC, accessed January 26, 2026, https://oac.cdlib.org/findaid/ark:%2F13030%2Fc82j6hc2
17. Vitamins and Beriberi | The Engines of Our Ingenuity - University of Houston, accessed January 26, 2026, https://engines.egr.uh.edu/episode/567
18. The Structural and Biochemical Foundations of Thiamin Biosynthesis - PMC, accessed January 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC6078420/
19. Celebrating Kazimierz Funk's 140th Birthday. The man who discovered vitamins, accessed January 26, 2026, https://researchinpoland.org/news/celebrating-kazimierz-funks-140th-birthday-the-man-who-discovered-vitamins/
20. The Agricultural Chemist at the Table: Land Grant Colleges, Experiment Stations, and the Birth of Nutrition Science in the United States, 1887 - UC Berkeley, accessed January 26, 2026, https://escholarship.org/content/qt1j32g7s7/qt1j32g7s7.pdf
21. Plain Text UTF-8 - Project Gutenberg, accessed January 26, 2026, https://www.gutenberg.org/ebooks/7983.txt.utf-8
22. Casimir Funk - Wikipedia, accessed January 26, 2026, https://en.wikipedia.org/wiki/Casimir_Funk
23. Casimir Funk - ISOM - International Society for Orthomolecular Medicine, accessed January 26, 2026, https://isom.ca/profile/casimir-funk/
24. Casimir Funk Collection - Online Archive of California, accessed January 26, 2026, https://oac4.cdlib.org/findaid/ark:/13030/c82j6hc2/entire_text/
25. Alternative Medicine Journal of Evidence-Based Complementary & - Korsakov Kenniscentrum, accessed January 26, 2026, https://korsakovkenniscentrum.nl/storage/uploads/2019/12/2011-fattal-thiamin-vitamin-b1.pdf
26. Thiamine - Wikipedia, accessed January 26, 2026, https://en.wikipedia.org/wiki/Thiamine

27. CHAPTER 1: Historical Context of Vitamin B - Books - The Royal Society of Chemistry, accessed January 26, 2026, https://books.rsc.org/books/edited-volume/1298/chapter/1178498/Historical-Context-of-Vitamin-B
28. (PDF) The Discovery of the Vitamins - ResearchGate, accessed January 26, 2026, https://www.researchgate.net/publication/241698057_The_Discovery_of_the_Vitamins
29. Vitamin B3 (Niacin) - Molecule of the Month - January 2019 (HTML version), accessed January 26, 2026, https://www.chm.bris.ac.uk/motm/vitaminB3/vitaminb3h.htm
30. Kazimierz (Casimir) Funk – Pioneer in Vitaminology – 101 Anniversary of His Discovery – Special Note Stanisław Berger* - SGGW, accessed January 26, 2026, https://ptnz.sggw.edu.pl/wp-content/uploads/2017/07/63_04_Berger.pdf
31. UCSF - UC San Francisco Electronic Theses and Dissertations - eScholarship, accessed January 26, 2026, https://escholarship.org/content/qt2g51w0qk/qt2g51w0qk.pdf
32. Casimir Funk: His Discovery of the Vitamins and... : Annals of ... - Ovid, accessed January 26, 2026, https://www.ovid.com/journals/anumds/fulltext/10.1159/000319165~casimir-funk-his-discovery-of-the-vitamins-and-their